

Wireless Communication Exam Notes

EX 715 (BEI) / EX 751 (BEX) · IV-I · TU, IOE · tiers over 22 papers, 2070–2082 BS

2. Equalization and Diversity Techniques

4 hrs · asked in **21 of 22 papers** · 7 syllabus sub-topics · 1 numerical family → Ch 5 numerical companion.

Read this first:

- **Equalization** **TOP 15/22** and **diversity** **TOP 11/22** are two of the three most reliable questions in the whole paper. **RAKE** **TOP 6/22** is close behind.
- Open every answer here with the **same framing sentence**: equalization, diversity and channel coding are three techniques used **independently or in tandem** to improve receiver signal quality.
 - **Equalization** undoes **ISI** from a time-dispersive channel.
 - **Diversity** undoes **fading** by giving the receiver more than one look at the signal.
 - **Channel coding** (Ch 6) adds **redundancy** so faded bits can be recovered.
- **Learn which impairment each one fixes.** Papers repeatedly ask “why is X needed”, and the mark is for naming the impairment, not describing the block diagram.
- **No numerical question on this chapter has ever appeared in a PYQ**, but the MSE optimum-weight derivation **HOT 3/22** is examined as bookwork, and the tutorial set carries diversity SNR problems. Both are in the companion.

5.1 Equalization: why, and how it adapts

Why is equalization needed? Describe the signal processing operation that minimizes ISI **2+6** **2+5** **4**
2+1+4

TOP 6/22 72 Ma, 73 Ma, 75 Bh, 73 Bh, 82 Bh, 80 Ba · **the highest-frequency question in the chapter**

Why it is needed

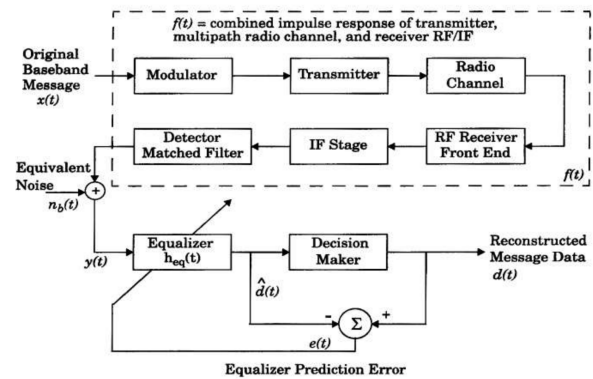
- A mobile channel is **time dispersive**: multipath copies arrive at different delays.
- A delayed copy of one symbol lands **on top of the next symbol** → **intersymbol interference (ISI)**.
- ISI happens whenever signal bandwidth W exceeds the coherence bandwidth B_C , ie whenever the channel is **frequency selective**.
- ISI is **the single largest bottleneck on high-rate data over a mobile channel**, and raising transmit power does not help: it raises the interference too.

Definition

- **Equalization** = any signal processing operation that **minimizes ISI**.
- The equalizer is the **inverse filter of the channel**. If the channel response is $f(t)$, the equalizer response $h_{eq}(t)$ satisfies

$$f(t) * h_{eq}(t) = \delta(t)$$

- So it **flattens** the frequency response, hence the name.
- Implemented at **baseband or at IF** in the receiver, where the complex envelope is available.
- **The mobile channel is random and time varying**, so a fixed filter cannot work. The equalizer must track the channel → it must be **adaptive**.



Block diagram of a simplified communications system using an adaptive equalizer at the receiver.

A communication system w/ an adaptive equalizer at the receiver.

$f(t)$ is the combined impulse response of transmitter, radio channel and receiver RF/IF stages. The equalizer inverts all three at once.

The error-driven loop

- Equalizer output $\hat{d}(t)$ is compared against a desired signal $d(t)$.
- Difference = **error signal** $e(t)$, fed back to retune the filter.
- If $d(t)$ is **not** fed back to adapt the filter, the equalizer is **linear**.
- If $d(t)$ is fed back to change subsequent outputs, it is **nonlinear**. **That single sentence is the definition of the linear / nonlinear split.**

Explain the training and tracking modes of an adaptive equalizer 4 1+3

HOT 3/22 78 Ch, 73 Bh, 81 Ba

Training mode (measure the channel)

- Transmitter sends a **known, fixed-length training sequence** immediately before the user data.
- Receiver **knows** that sequence, so it can compute the error exactly.
- The adaptive algorithm runs on this sequence and drives the filter taps to their optimum values, minimising BER.
- Typical sequence: a **pseudo-random binary signal** or a fixed bit pattern.
- Design rule: the training sequence must be **long enough** for the equalizer to converge under the **worst possible** channel conditions.
- In GSM the training sequence sits in the **middle of every burst** (the midamble), so the estimate is equally fresh at both ends of the burst. verified/added

Tracking mode (follow the channel)

- Starts the instant training ends and **user data** begins.
- There is no known reference now, so the algorithm uses the **detected symbols** as the reference (decision directed).
- It keeps **tracking** the channel, which is changing as the mobile moves.
- **The whole point: a mobile channel changes continuously, so a one-off training would go stale within milliseconds.**

Three factors that set the time span an equalizer can handle

- **Length of the training sequence** and the **rate of convergence** of the algorithm.
- **Speed of the mobile**, which sets the fading rate, the Doppler spread and hence the coherence time.
- **Number of taps**, which must cover the maximum expected **delay spread**. More taps → more delay spread covered, but **circuit complexity and processing time rise with tap count**.

For the MSE algorithm, determine the optimum solution for the weights 8 7

HOT 3/22 79 Bh, 79 Ch, 80 Ch · asked as a full derivation w/ diagram → worked in the Ch 5 companion

Structure of the adaptive equalizer + Fig

- A **transversal filter** w/ N delay elements, $N + 1$ taps and $N + 1$ tunable **complex weights** $w_{0k} \dots w_{Nk}$.
- Subscript k = discrete time index, and it appears on the weights to show they are **updated sample by sample**.
- Single input y_k at each instant. Its value depends on the instantaneous state of the radio channel and on the noise.
- Weights are updated continuously by the **adaptive algorithm**, which is driven by the error signal e_k .

Setting up the cost function

- Input vector and weight vector:

$$\mathbf{y}_k = [y_k \ y_{k-1} \ \dots \ y_{k-N}]^T, \quad \mathbf{w}_k = [w_{0k} \ w_{1k} \ \dots \ w_{Nk}]^T$$

- Equalizer output $\hat{d}_k = \mathbf{y}_k^T \mathbf{w}_k$.
- **Error signal**, w/ x_k the known transmitted symbol:

$$e_k = d_k - \hat{d}_k = x_k - \mathbf{y}_k^T \mathbf{w}_k$$

- Squaring:

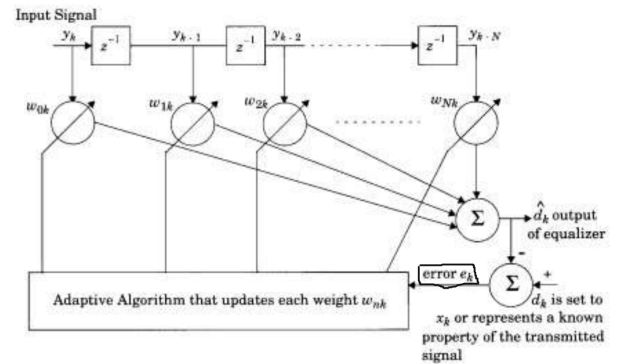
$$e_k^2 = x_k^2 + \mathbf{w}_k^T \mathbf{y}_k \mathbf{y}_k^T \mathbf{w}_k - 2x_k \mathbf{y}_k^T \mathbf{w}_k$$

- **Mean square error** is the expectation:

$$\xi = E[e_k^2] = E[x_k^2] + \mathbf{w}^T \mathbf{R} \mathbf{w} - 2\mathbf{p}^T \mathbf{w}$$

where

$$\mathbf{R} = E[\mathbf{y}_k \mathbf{y}_k^T] \text{ (input correlation matrix),} \quad \mathbf{p} = E[x_k \mathbf{y}_k] \text{ (cross correlation)}$$



A basic linear equalizer during training.

Basic linear equalizer during training. d_k is the known training symbol, e_k the error that drives the adaptive algorithm.

Minimising it

- ξ is a **quadratic** function of the weights, so its surface is a **bowl-shaped paraboloid** (a hyperparaboloid for more than 2 taps).
- It is **concave upward**, so a **single global minimum** exists. **That is why gradient descent cannot get stuck here.**
- Take the gradient w.r.t. the weight vector:

$$\nabla \xi = \left[\frac{\partial \xi}{\partial w_0} \ \frac{\partial \xi}{\partial w_1} \ \dots \ \frac{\partial \xi}{\partial w_N} \right]^T = 2\mathbf{R}\mathbf{w} - 2\mathbf{p}$$

- Set $\nabla \xi = 0$:

$$\hat{\mathbf{w}} = \mathbf{R}^{-1}\mathbf{p}$$

valid as long as **R** is **non-singular**, ie has an inverse.

- Substituting back gives the **minimum MSE**:

$$\xi_{\min} = E[x_k^2] - \mathbf{p}^T \hat{\mathbf{w}} = E[x_k^2] - \mathbf{p}^T \mathbf{R}^{-1} \mathbf{p}$$

- **Minimising the MSE tends to minimise the bit error rate**, which is why MSE is the standard cost function even though BER is what actually matters.

The iterative form

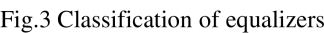
- Direct matrix inversion is expensive, so in practice the weights are stepped:

$$\text{new weights} = \text{previous weights} + \mu \times (\text{previous error}) \times (\text{current input vector})$$

- μ is a **step size** adjusted to control how far the weights move per iteration.
- Repeated rapidly in a loop until **convergence**, at which point the weights are frozen until the next training sequence.

Explain the structure of the linear transversal equalizer, DFE and MLSE 6+1+4 4 6+1+5

Equalization Techniques



delay elements

$y(t) + n_b(t)$ → T_s — T_s — T_s — T_s —

clock with period τ —

Taps

The diagram illustrates a discrete-time adaptive filter structure. The input signal $y(t)$ is fed into a summation junction Σ . This junction also receives feedback signals from four feedback taps, labeled b_1, b_2, b_3, b_4 , which are connected to the output of the filter at four different stages. The output of the first summation junction is passed through a series of four delay blocks, each labeled T_s . After the fourth delay block, the signal is fed into a second summation junction Σ . This second junction also receives feedforward signals from four feedforward taps, labeled a_1, a_2, a_3, a_4 , which are connected to the output of the filter at four different stages. The output of the second summation junction is the estimated signal $\hat{d}(t)$.

Tapped delay line filter with *both* feedforward and feedback taps.

Linear equalizers

- Output is **not** fed back to change subsequent outputs.
- Implemented as an **FIR filter**: current and past values of the received signal are linearly weighted by the tap coefficients and summed.
- If the taps are analog, the output is **sampled at the symbol rate** before the decision device.
- **Linear transversal equalizer (LTE)** + Fig: a tapped delay line, delay T_s per element, transfer function a function of z^{-1} (or $\exp(-j\omega T_s)$).
- The simple LTE uses **feedforward taps only**. A version w/ both feedforward and feedback taps exists but is **unstable and rarely used**.
- **Lattice equalizer**: same job, but built as a lattice filter.
 - (+) numerically **stable**, fast convergence.
 - (+) the **order can be changed dynamically** without restarting, so a short filter can be used on a benign channel and a long one on a bad channel.
 - (-) structurally more complicated.
- **(-) Linear equalizers perform badly on channels with deep spectral nulls**, because inverting a null means enormous gain at that frequency, which also amplifies the noise. That limitation is the whole reason nonlinear equalizers exist.

Nonlinear equalizers 4

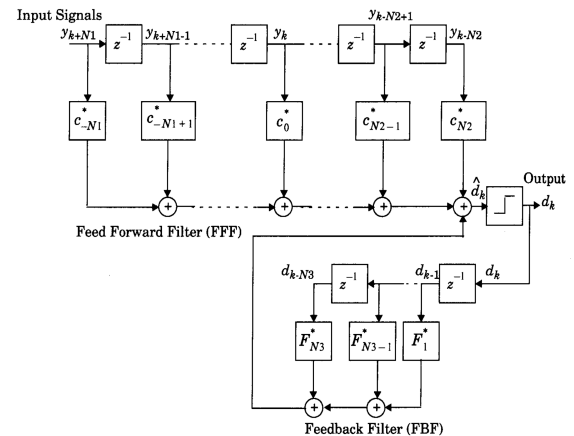
- Used when the channel distortion is **too severe for a linear equalizer**.
- Three effective methods: **DFE, maximum likelihood symbol detection, MLSE**.

Decision feedback equalizer (DFE)

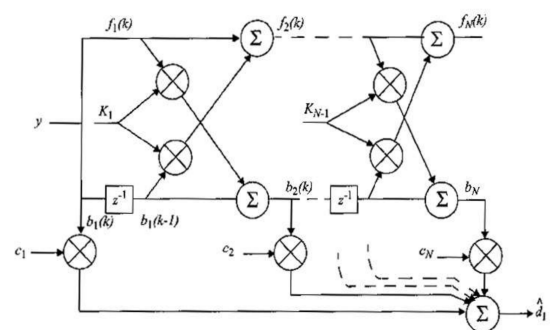
- Basic idea:** once a symbol has been detected and **decided upon**, the ISI it will cause on **future** symbols can be **estimated and subtracted out** before those symbols are detected.
- Two filters:
 - Feedforward filter (FFF):** an ordinary linear transversal filter on the received signal.
 - Feedback filter (FBF):** driven by **previously detected symbols**, and its output is subtracted.
- Output:

$$\hat{d}_k = \sum_{n=-N_1}^{N_2} c_n^* y_{k-n} + \sum_{i=1}^{N_3} F_i d_{k-i}$$

- (+) The FBF removes ISI without amplifying noise**, because it works on already decided symbols, which are noise free. That is exactly what a linear equalizer cannot do.
- (-) Error propagation:** one wrong decision feeds a wrong correction into every following symbol.
- Realised in **direct transversal** form or as a **lattice** filter.
- Predictive DFE** (Belfiore and Park): the FBF acts as a **noise predictor** instead. Performs as well as the conventional DFE in the limit of many taps, and the **RLS** algorithm gives it fast convergence.



Decision feedback equalizer: feedforward filter on the received signal, feedback filter driven by detected symbols.

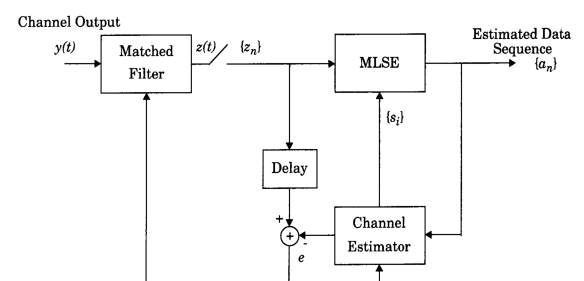


Lattice equalizer structure.

Maximum likelihood sequence estimation (MLSE) 6 4

HOT 4/22 71 Ma, 72 Ma, 72 Ash, 81 Bh · always asked with a block diagram

- Ordinary equalizers decode **each symbol on its own**. MLSE instead **tests all possible data sequences** and picks the sequence with the **maximum likelihood** of having been sent.
- It is optimal in the sense of minimising the probability of a sequence error**. No equalizer can do better.
- First proposed by **Forney**, who implemented the MLSE estimator using the **Viterbi algorithm** (see Ch 6).
- Three blocks + Fig:**
 - Adaptive matched filter** matched to the combined channel, giving $z(t)$.
 - MLSE** proper, running the Viterbi search over the trellis.
 - Channel estimator** in a feedback loop, keeping the filter and the metric matched to the current channel.
- Requirements:**
 - knowledge of the **channel characteristics**, to compute the branch metrics,
 - knowledge of the **statistical distribution of the noise**.
- (-) Huge computational load:** complexity grows **exponentially** w/ the channel delay spread in symbols, so it is only practical when the delay spread covers a small number of symbols. GSM is exactly that case.



MLSE with an adaptive matched filter and a channel estimator in the loop.

Explain any two adaptive equalization algorithms 4 5

73 Ma, 74 Ma · name all three, then detail two

	Zero forcing (ZF)	Least mean square (LMS)	Recursive least squares (RLS)
Criterion	Force ISI to zero at the sampling instants	Minimise the mean square error	Minimise a weighted sum of past squared errors
Update	Taps set so the combined response is 1 at $t = 0$ and 0 at every other kT_s	$\omega_{k+1} = \omega_k + \mu e_k \mathbf{y}_k^*$	Uses the Kalman gain vector and an inverse correlation matrix
Complexity	Lowest	Low, $2N + 1$ operations per iteration	High , $\propto N^2$
Convergence	Immediate (non-adaptive form)	Slow , needs a long training sequence	Fastest , an order of magnitude fewer iterations
Noise	Amplifies noise badly at spectral nulls	Well behaved	Well behaved
Use it when	The channel is benign and simple hardware matters	Convergence speed is not critical	The channel changes fast, or training must be short

- **Zero forcing in more detail:** it is the **inverse filter** of the channel, $H_{eq}(f) = 1/H_{ch}(f)$. It removes ISI completely **in a noise-free channel**.
- **(–) In reality it does not perform well:** where the channel has a spectral null, $1/H_{ch}(f)$ becomes huge, so the equalizer applies enormous gain exactly where the noise is unopposed. It **amplifies noise** and can perform worse than no equalizer at all.
- **How to choose:** the **speed of the mobile** sets the fading rate and Doppler spread, hence the coherence time. The algorithm must converge well inside that coherence time, so a fast-moving mobile forces RLS despite its cost.

5.3 Diversity

Why is diversity needed? Explain the different types of diversity technique 2+6 4+6 2+2 3+3+2 2

TOP 11/22 70 Bh, 71 Bh, 71 Ma, 73 Ma, 74 Bh, 74 Ma, 76 Bh, 78 Ch, 80 Ba, 80 Bh, 81 Ba · second most repeated question in the chapter

The concept (the “why” marks live here)

- **Objective:** combine multiple copies of the signal so as to **reduce the effect of deep fades**.
- **The core idea in one sentence:** if one radio path is in a deep fade, an **independent** path probably is not. Having more than one path to choose from improves both the instantaneous and the average SNR.
- Typical improvement: **20 to 30 dB**.
- (+) **Requires no training overhead**, unlike equalization.
- (+) Large link improvement for little added cost.
- Diversity decisions are made **entirely at the receiver** and are unknown to the transmitter.
- **Equalization fixes ISI. Diversity fixes fading. They solve different problems and are often used together.**

Microscopic Vs macroscopic diversity

- **Microscopic diversity** counters **small-scale** fading.
 - Selects the best signal at all times out of several rapidly fading ones.
 - Antennas separated by a fraction of a wavelength are enough.
- **Macroscopic diversity** counters **large-scale** fading, ie shadowing.
 - Uses **base stations that are not shadowed**, sufficiently separated in space.
 - Eg CDMA soft handoff, where two BSs serve one mobile.

Types of diversity (name all five)

- **Space (antenna) diversity**
 - Two or more **physically separated** receiving antennas.
 - Separation of $\lambda/2$ or more at the mobile gives essentially **uncorrelated envelopes**.
 - At a BS the separation must be much larger, **up to 20 wavelengths**, because the BS antenna is high and sees a narrow angular spread.
 - **The most common form, and the one the examiner means when the type is unspecified.**
- **Polarization diversity**
 - Send or receive the same information on **two orthogonal polarizations**, normally vertical and horizontal.
 - Relies on the **decorrelation of the two receive ports**, so the two ports must stay cross-polarized.
 - (+) **Needs no antenna spacing at all**, so one dual-polarized antenna replaces two widely spaced ones. Huge saving on BS tower space.
 - (–) Only 2 branches are possible, and each port gets about **3 dB less power** than a single co-polarized antenna.

Diversity technique

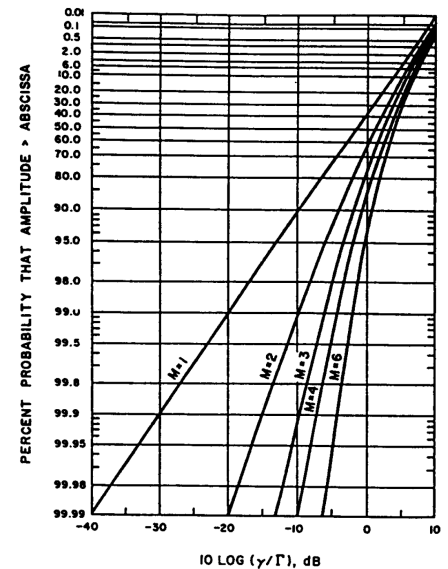


Fig. 11 Graph of probability distributions of $SNR = \gamma/\bar{\gamma}$

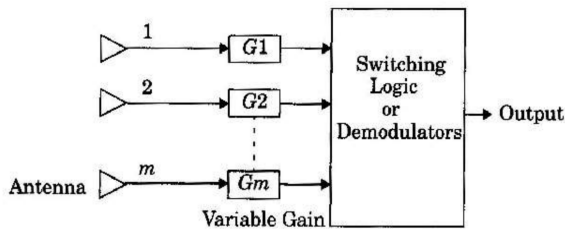
Probability distribution of SNR threshold for M -branch selection diversity. $\bar{\gamma}$ is the mean SNR on each branch. **Each extra branch pushes the low-SNR tail sharply to the right, which is the diversity gain.**

- **Frequency diversity**
 - Transmit the same information on **more than one carrier frequency**.
 - Carriers separated by **more than the coherence bandwidth** do not fade together.
 - (–) Costs extra spectrum and an extra transmitter chain.
 - Used on microwave LOS links, where it doubles as equipment redundancy.
- **Time diversity**
 - Repeatedly transmit the information at time spacings **exceeding the coherence time** of the channel.
 - Repetitions then fade independently.
 - **Two implementations (72 Ash asks for exactly two, 5):**
 - **Interleaving with channel coding:** spread a codeword's bits far apart in time so a burst error becomes several correctable single errors.
 - **RAKE receiver** in CDMA: multipath copies are themselves time-shifted versions of the same signal, so combining them **is time diversity for free**.
 - **ARQ** retransmission is a third, cruder form.
 - (–) Costs delay, which **voice cannot tolerate beyond about 40 ms**.

verified/added

Explain the combining methods used for space diversity 4+4 3+3+2 2+2 4

TOP 7/22 74 Ma, 73 Ma, 74 Bh, 71 Ma, 78 Ch, 80 Ba, 80 Bh · 78 Bh asks specifically for **feedback Vs maximal ratio combining** · 80 Bh asks for MRC alone



Generalized block diagram for space diversity: m antennas, variable gains, then switching logic or a demodulator.

Space diversity reception falls into **four** categories. They differ only in **how the m branches are weighted and combined**, so answer them as one family. The general model has m branches whose gains are adjusted to give the same average SNR per branch.

Method	How it works	Cost
Selection	Monitor SNR on all m branches, connect the highest to the demodulator	m receivers, needs all channels monitored
Feedback / scanning	Scan branches in a fixed sequence , take the first above a preset threshold , keep it until it falls below	Only one receiver
Maximal ratio (MRC)	Co-phase all branches, weight each by its own signal-voltage to noise-power ratio , then sum	m receivers, needs accurate gain estimates
Equal gain (EGC)	Co-phase all branches, weight every one by unity , then sum	m receivers, no gain estimate

Selection diversity

- **Principle:** pick the best signal at every instant.
- Simplest in concept, but **requires knowledge of all channels** → m monitoring receivers.
- **Performance:** for M branches w/ mean SNR Γ each,

$$\Pr[\text{SNR} \leq \gamma] = \left(1 - e^{-\gamma/\Gamma}\right)^M$$

and the average SNR gain is

$$\frac{\bar{\gamma}_M}{\Gamma} = \sum_{k=1}^M \frac{1}{k}$$

- **Diminishing returns:** going from 1 to 2 branches gains $1.5\times$, 2 to 3 only $1.83\times$. **Most of the benefit is in the second branch.**

Feedback or scanning diversity

- **Principle:** the M signals are scanned in a **fixed sequence** until one is found **above a predetermined threshold**.
- That signal is used **until it falls below threshold**, then scanning restarts.
- (+) **Simple to implement, only one receiver is required.** That is its whole selling point.
- (–) The resulting fading statistics are **somewhat inferior** to the other three, because the branch in use is merely acceptable, not best.

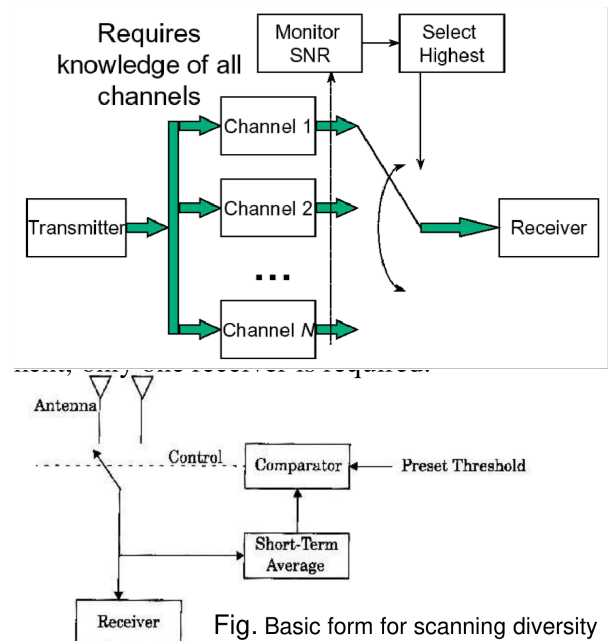


Fig. Basic form for scanning diversity

Top: selection diversity, monitor SNR then select highest.

Bottom: scanning diversity, one receiver plus a comparator against a preset threshold.

Maximal ratio combining (MRC) (80 Bh asks for this alone, 2)

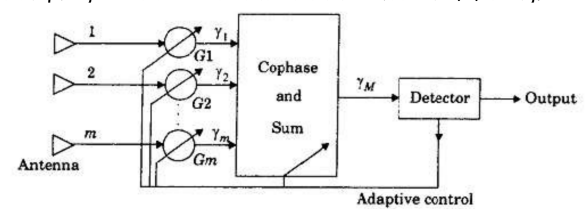
- **Principle:** combine **all** the signals, **co-phased** and **weighted**, so as to have the highest achievable SNR at all times.
- Branches are weighted by their **signal voltage to noise power ratio**, then summed.
- Co-phasing first is essential, otherwise the branches would partly cancel.
- **Output SNR is the sum of the individual SNRs:**

$$\gamma_M = \sum_{i=1}^M \gamma_i$$

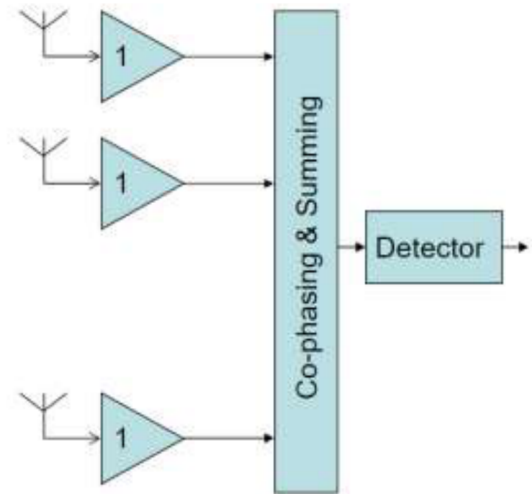
- **So MRC can produce an acceptable output even when none of the individual branches is acceptable on its own.** That is the line the examiner wants.
- (+) **Gives the best statistical reduction of fading** of any linear combiner.
- (–) Needs an **accurate estimate of each branch's amplitude gain**, so the receiver is the most complex of the four.

Equal gain combining (EGC)

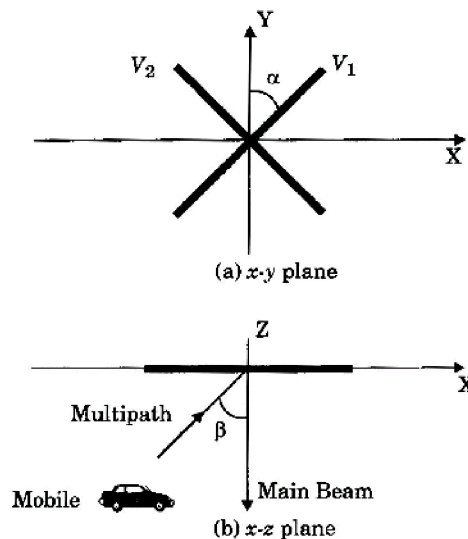
- Used when it is **not convenient** to provide the variable weighting MRC needs.
- Branch weights are **all set to unity**, but the signals are still **co-phased**.
- Lets the receiver exploit signals received simultaneously on every branch.
- **Performance is only marginally inferior to MRC, and superior to selection diversity**, for far less hardware. That trade is why EGC is common in practice.



Maximal ratio combiner: weight, co-phase, sum, then detect.



Equal gain combining: unity weights, co-phasing and summing only.

Diversity TechniquesTheoretical model for base station polarization diversity: two orthogonal receive ports at $\pm\alpha$ in the x - y plane.

5.4 RAKE receiver and interleaving

Explain the working of an M-branch RAKE receiver with a block diagram 8 3+5 4 3

TOP 6/22 70 Ma, 72 Ash, 73 Bh, 76 Bh, 77 Ch, 79 Bh · 73 Bh asks **how it exploits time diversity**, so say that explicitly

What it is

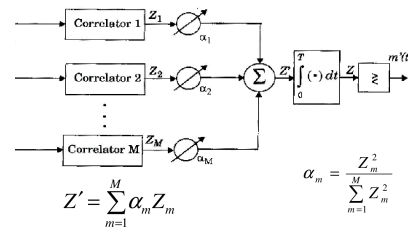
- A diversity receiver designed specifically for CDMA.
- In an ordinary receiver multipath is a problem. **In a RAKE receiver multipath is the resource.**
- It **collects the time-shifted versions** of the original signal by providing a **separate correlation receiver** for each multipath component.
- Each correlator is called a **finger**. Hence “rake”: fingers gathering energy that would otherwise be wasted.

Why it works only in CDMA (the key insight)

- The multipath components are **practically uncorrelated** from one another once their relative propagation delays exceed **one chip period**.
- That comes from the **low autocorrelation** of the PN spreading sequence: a copy delayed by more than a chip correlates almost to zero.
- So a correlator locked to delay τ_m sees component m clearly and treats the others as noise.
- **A narrowband system cannot do this** because its copies are not separable in delay.

How it exploits time diversity (73 Bh, 3)

- Time diversity means receiving the same information at times separated by more than the channel's coherence time or, here, resolvable in delay.
- The multipath copies **are** exactly that: the same symbol, arriving at different delays, faded **independently**.
- So the RAKE gets time diversity **without spending any extra bandwidth, power or time**, which no other diversity scheme manages.



(finger) RAKE receiver implementation. Each correlator detects a time shifted

An M -branch (M -finger) RAKE. Each correlator detects one time-shifted version of the CDMA transmission, delayed by at least one chip from the others.

Operation, step by step

- **Multiple correlators** separately detect the M **strongest** multipath components. Correlator m is delay-locked to component m and outputs Z_m .
- Outputs are **weighted** by α_m to produce a better estimate than any single component could.
- **Combine** (this is MRC applied to multipath):

$$Z' = \sum_{m=1}^M \alpha_m Z_m, \quad \alpha_m = \frac{Z_m^2}{\sum_{m=1}^M Z_m^2}$$

- **Demodulation and bit decision** are then made on the weighted sum Z' , not on any one finger.
- If one finger is in a deep fade, the others carry the symbol.
- **A CDMA link in a rich multipath environment can therefore outperform the same link in a “clean” one-path environment.**

What is interleaving? Explain its working principle 4 2+6

HOT 3/22 80 Bh, 81 Bh, 81 Ba

Why it is needed

- A fade does not corrupt bits at random. It wipes out a **burst** of consecutive bits.
- Channel codes (Ch 6) are designed to fix **scattered** errors. A burst overwhelms them.
- **Interleaving scrambles the time order of the source bits before channel coding**, so that a burst on the channel arrives at the decoder as **isolated single errors spread across many codewords**, which the code can fix.
- Speech coders make this worse: several “**important**” bits occur in succession, so without interleaving one fade can destroy an entire speech frame.
- **Interleaving is time diversity implemented in the digital domain.**

Block interleaver +Fig

- Formats the encoded data into a rectangular array of m rows and n columns, interleaving nm bits at a time.
- Each row usually holds one n -bit source word.
- An interleaver of **degree (depth) m** has m rows.
- **Write in by column**: bits are placed by increasing the row number for each successive bit, filling the columns.
- **Read out by row**, and transmit.
- Effect: **original neighbouring bits end up m bit periods apart**.
- At the receiver the **de-interleaver** does the inverse: store by column, clock out one row at a time.
- So an (n, k) code that can handle a burst of $b < (n - k)/2$ can be combined w/ an interleaver of degree m to make an (mn, mk) block code that handles bursts of mb bits.

Convolutional interleaver

- Same job, but built as a bank of shift registers of increasing length rather than a rectangular array.
- Can replace a block interleaver in the same fashion, and is **ideally suited for use with convolutional codes**.
- (+) Roughly **half the delay and half the memory** of a block interleaver of the same effective depth. verified/added

The cost

- **Delay**, and the delay grows with interleaver depth.
- **Human speech tolerates delays below about 40 ms**, which caps the depth a voice system can use. Data services can interleave far more deeply.

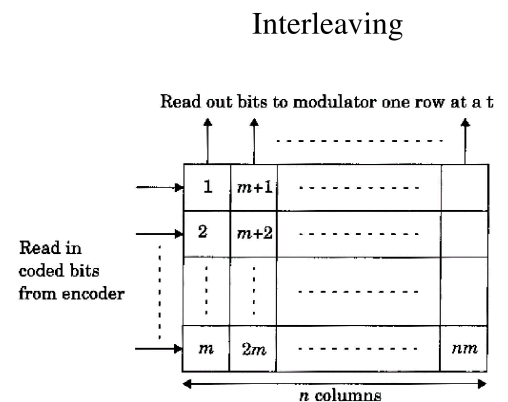


Fig. 17 Block interleaver where source bits are read into columns and

Block interleaver: coded bits are read in by column and read out to the modulator one row at a time.

Abbreviations used in these notes

Only these are ever shortened. Anything not on this list is written out in full.

Short	Means	Short	Means
BS	base station	BTS	base transceiver station
MS	mobile station	MSC	mobile switching centre
BSC	base station controller	MSK	minimum shift keying
CCI	co-channel interference	ACI	adjacent channel interference
SIR, S/I	signal to interference ratio	SNR, S/N	signal to noise ratio
GoS	grade of service	MAHO	mobile assisted handoff
FCA	fixed channel assignment	DCA	dynamic channel assignment
LOS	line of sight	NLOS	non line of sight
FDMA	frequency division multiple access	TDMA	time division multiple access
CDMA	code division multiple access	OFDM	orthogonal frequency division multiplexing
FDD	frequency division duplex	TDD	time division duplex
GSM	global system for mobile communications	AMPS	advanced mobile phone system
PCS	personal communication system	DAS	distributed antenna system
BCCH	broadcast control channel	EIRP	effective isotropic radiated power
ERP	effective radiated power	RF	radio frequency
Tx, TX	transmitter	Rx, RX	receiver
ISI	intersymbol interference	PDP	power delay profile
DFE	decision feedback equalizer	MLSE	maximum likelihood sequence estimation
LMS	least mean square	RLS	recursive least squares
MSE	mean square error	MMSE	minimum mean square error
ZF	zero forcing	MRC	maximal ratio combining
EGC	equal gain combining	BER	bit error rate
PCM	pulse code modulation	DPCM	differential PCM
ADPCM	adaptive differential PCM	LPC	linear predictive coder
RPE-LTP	regular pulse excited, long term prediction	CELP	code excited linear prediction
RELp	residual excited linear prediction	MPE	multipulse excited
VSELP	vector sum excited linear prediction	ATC	adaptive transform coding
SBC	sub-band coding	DCT	discrete cosine transform
FEC	forward error correction	ARQ	automatic repeat request
BCH	Bose–Chaudhuri–Hocquenghem code	RS	Reed–Solomon code
TCM	trellis coded modulation	PN	pseudo-noise

Short	Means	Short	Means
MIMO	multiple input multiple output	OFDMA	orthogonal FDM access
QAM	quadrature amplitude modulation	PSK	phase shift keying
GMSK	Gaussian minimum shift keying	IMT	international mobile telecommunications
ASK	amplitude shift keying	FSK	frequency shift keying
BPSK	binary phase shift keying	QPSK	quadrature phase shift keying
OQPSK	offset (staggered) QPSK	DPSK,	differential (binary) PSK
		DBPSK	
CPFSK	continuous phase FSK	MFSK,	M-ary FSK, M-ary PSK
		MPSK	
DSSS	direct sequence spread spectrum	FHSS	frequency hopped spread spectrum
SS, SSM	spread spectrum (modulation)	PG	processing gain
PAPR	peak-to-average power ratio	CP	cyclic prefix
FFT, IFFT	(inverse) fast Fourier transform	ICI	inter-carrier interference
PSD	power spectral density	CFO, SFO	carrier, sampling frequency offset
NRZ, RZ	non-return, return to zero	AMI	alternate mark inversion
AM, FM, PM	amplitude, frequency, phase modulation	DSB, SSB, VSB	double, single, vestigial sideband
S/P, P/S	serial-parallel converter	D/A, A/D	digital-analog converter
LPF, BPF	low pass, band pass filter	ISM	industrial, scientific, medical band
SC-FDMA	single carrier FDMA	<i>BT</i>	GMSK filter bandwidth × symbol time
PSTN	public switched telephone network	VoIP	voice over IP
FVC, RVC	forward, reverse voice channel	FCC, RCC	forward, reverse control channel
SSMA	spread spectrum multiple access	SDMA	space division multiple access
FHMA	frequency hopped multiple access	DSMA	direct sequence multiple access (= CDMA)
FCDMA	hybrid FDMA/CDMA	DS/FHMA	hybrid direct sequence / frequency hopped
TCDMA	time division CDMA	TDFH	time division frequency hopping
IM	intermodulation (product)	MU-MIMO	multi-user MIMO
WLAN	wireless local area network	WMAN	wireless metropolitan area network
WPAN	wireless personal area network	AP	access point
BSS, IBSS	basic service set, independent (ad hoc) BSS	ESS	extended service set
CSMA/CA	carrier sense multiple access with collision avoidance	CSMA/CD	... with collision detection
DCF	distributed coordination function	PCF	point coordination function
RTS/CTS	request to send / clear to send	CCK	complementary code keying
WEP, WPA	wired equivalent privacy, WiFi protected access	U-NII	unlicensed national information infrastructure band
IoT	internet of things	mmWave	millimetre wave
eMBB	enhanced mobile broadband	URLLC	ultra-reliable low latency comm.
mMTC	massive machine type comm.	NOMA	non-orthogonal multiple access
D2D	device to device	C-RAN	cloud radio access network
PCN	personal communication network	SMS	short message service
attr(s)	attribute(s)	no.	number
freq	frequency	info	information
ops	operations	config	configuration
rm	remove	vs, Vs	versus
w/, w/o	with, without	Eg	example / for example
+Fig	draw a figure	(I)	importance / need / significance
→	leads to / therefore	+	and

Standard and system names (proper names, kept as printed): **AMPS** advanced mobile phone system · **ETACS** extended total access communication system · **NMT** Nordic mobile telephone · **NTT** Nippon Telegraph and Telephone · **IMTS** improved mobile telephone service · **PDC** Pacific digital cellular · **IS-95** cdmaOne · **IS-136** digital AMPS · **HSCSD** high speed circuit switched data · **GPRS** general packet radio service · **EDGE** enhanced data rates for GSM evolution · **W-CDMA** wideband CDMA · **UMTS** universal mobile telecommunications system · **cdma2000 1xRTT / 1xEV-DO / 1xEV-DV** the cdma2000 3G family · **LTE, LTE-A** long term evolution (advanced) · **WiMAX** IEEE 802.16 · **WiFi** IEEE 802.11 · **DECT** digital enhanced cordless telecommunications · **CT2** cordless telephone 2 · **PHS** personal handyphone system · **USDC** US digital cellular (IS-136) · **JDC** Japanese digital cellular (later PDC) · **Bluetooth** IEEE 802.15.1 · **Zigbee** IEEE 802.15.4 · **PACS** personal access communications system · **POCSAG, FLEX** paging protocols · **CDPD** cellular digital packet

data · **Mobitex** a dedicated packet data system · **TETRA** terrestrial trunked radio · **DAB** digital audio broadcasting · **DVB-T** digital video broadcasting, terrestrial · **ADSL** asymmetric digital subscriber line · **5G NR** 5G new radio.

One clash to watch: **FCC** is the *forward control channel* throughout these notes. The US regulator is always written out as the Federal Communications Commission.

Exam-paper month codes (used in the year tags): Ba = Baishakh · Bh = Bhadra · Ash = Ashwin · Ma = Magh · Ch = Chaitra.

Bold year tags are Regular exams, plain are Back exams. Monospace year tags are BEI (EX 715); the rest are BEX (EX 751).

Tier chips over 22 papers, 2070–2082 BS: **TOP /22** = topic asked in 6 or more papers · **HOT /22** = 3 to 5 papers · **PIN /22** = 1 or 2 papers but worth ≥ 6 marks. **Mark chips** such as **2+5** are the marks that question actually carried. **syllabus, no PYQ yet** marks a syllabus topic that has never been examined but is written up anyway. **verified/added** marks anything researched and added that was not in the source notes.